

FILTER DESIGN TEST CASE

Duplexer Module - Band 26 (850 MHz) for IoT Narrowband Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Duplexer Module filter targeting Band 26 (850 MHz) for IoT Narrowband Module. The filter is designed on PZT on Si substrate with a target center frequency of 5230.4 MHz and bandwidth of 273.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the IoT Narrowband Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Duplexer Module
- Frequency Band: Band 26 (850 MHz)
- Application: IoT Narrowband Module
- Substrate: PZT on Si
- Package: WLP (Wafer Level Package)
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: Newbury Park Design Center

This specification applies to all variants of the Duplexer Module design targeting Band 26 (850 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5230.4 MHz
Bandwidth (3 dB):	273.1 MHz
Insertion Loss Target:	≤ 2.0 dB
Return Loss Target:	≥ 11.0 dB
Out-of-Band Rejection:	≥ 32.0 dB
Isolation (Duplexer):	≥ 41.0 dB
Q Factor:	11997
Temperature Coefficient:	-10.3 ppm/°C
Die Size:	1.9 x 1.0 mm
Wafer Size:	6-inch
Number of Resonators:	4
Electrode Material:	Al

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the Duplexer Module template project.
2. Define the substrate stack: PZT on Si with specified layer thicknesses.
3. Set design targets: center freq 5230.4 MHz, BW 273.1 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.0 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (WLP (Wafer Level Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, PZT on Si).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 15691 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.0 dB
- Return loss in passband shall be >= 11.0 dB
- Out-of-band rejection at +/- 410 MHz offset >= 32.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 6.7 MHz
- Power handling shall be >= +30 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 86%

6. TEST RESULTS

Measurement Date: 2024-05-18
Devices Tested: 65
Insertion Loss (meas): 1.68 dB
Return Loss (meas): 13.2 dB
Rejection (meas): 34.5 dB
Isolation (meas): 45.9 dB
Q Factor (meas): 11997
Group Delay Variation: 5.9 ns
Power Handling: +26 dBm
Die Yield: 88.4%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Duplexer Module design demonstrated acceptable performance for Band 26 (850 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Simulation-to-measurement correlation was within 3% for all key parameters.
- e) No delamination observed after 1000-cycle thermal shock test.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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